ESERCITAZIONE 02

⑤ $f(x) = \sqrt{(x-3)(x-7)(x-11)}$

D:

$(x-3)(x-7)(x-11) \geq 0$

$x-3 \geq 0 \rightarrow x \geq 3$

$x-7 \geq 0 \rightarrow x \geq 7$

$x-11 \geq 0 \rightarrow x \geq 11$



$3 \leq x \leq 7 \ \vee \ x \geq 11$

$[3,7] \cup [11,+\infty)$

$E = \{x \in \mathbb{R}: 3 \leq x \leq 7 \text{ e } x \geq 11\}$

$g(x) = \frac{1}{\sqrt{x-2} + \sqrt{x-8}}$

D:

$\sqrt{x-2} + \sqrt{x-8} \neq 0$ ← Sempre verificata

$x-2 \geq 0 \rightarrow x \geq 2$

$x-8 \geq 0 \rightarrow x \geq 8$ ⟶ Più restrittiva

$[8, +\infty)$

DOMANDA
PERCHÈ NON SI FA LO STUDIO DEL SEGNO E SI PRENDE DIRETTAMENTE $[8,+\infty)$?

PERCHÈ SI PRENDE LA PARTE DELLO STUDIO DEL SEGNO DOVE LE 2 PARTE COESISTONO? E NON SI FA LO STUDIO DEL SEGNO COME SEMPRE?

$h(x) = \log_3(-x^2+11x-28)$

D:

$-x^2+11x-28 > 0$

$x^2-11x+28 < 0$

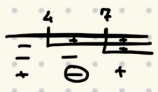
$\Delta = 9$

$x_{1,2} = \frac{11 \pm \sqrt{9}}{2} = \frac{11 \pm 3}{2} = 7, 4$

$1(x-7)(x-4) > 0$

$x-7 > 0 \rightarrow x > 7$

$x-4 > 0 \rightarrow x > 4$



$4 < x < 7$

$E = (4,7)$

$E = \{x \in \mathbb{R}: 4 < x < 7\}$

$K(x) = \sqrt[6]{\cos(3x) - 1/2}$

D:

$\cos(3x) - 1/2 \geq 0$

$\cos(3x) \geq 1/2$

$\cos t \geq 1/2 \qquad t = 3x$

$\cos(\pi/3) = \cos(-\pi/3) = 1/2$



$t \in [-\pi/3, \pi/3] \cup [-\pi/3 + 2\pi, \pi/3 + 2\pi]$

$\bigcup_{k \in \mathbb{Z}} [-\pi/3 + 2k\pi, \pi/3 + 2k\pi]$

$E = \{x \in \mathbb{R}: \cos(3x) \geq 1/2\} = \bigcup_{k \in \mathbb{Z}} [-\pi/9 + 2/3 k\pi, \pi/9 + 2/3 k\pi]$

PERCHÈ MOLTIPLICHIAMO?

$h(x) = f(x) = \frac{\sin(2x)}{\tan(x) - 1}$

D:

$\tan(x) - 1 \neq 0$

$\tan(x) \neq 1$

$x \neq \pi/4 + k\pi \quad k \in \mathbb{Z}$

COME FACCIO A CAPIRE CHE $x \neq 45° \ x \neq \pi/4$?

$E = \mathbb{R} \setminus \{\pi/4 + k\pi : k \in \mathbb{Z}\}$

⑥ $g(x) = \sqrt{6^{\sqrt{x-6}} - 36}$

D:

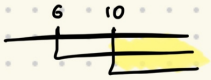
$6^{\sqrt{x-6}} - 36 \geq 0 \longrightarrow x-6 \geq 0 \rightarrow x \geq 6$

$6^{\sqrt{x-6}} \geq 6^2$

$\sqrt{x-6} \geq 2$

$x-6 \geq 4$

$x \geq 10$



$x \geq 10$

$E = [10, +\infty)$

$E = \{x \in \mathbb{R}: x \geq 10\}$

$h(x) = \sqrt{x-6} + \sqrt{36-x^2}$

D:

$x-6 \geq 0 \rightarrow x \geq 6$

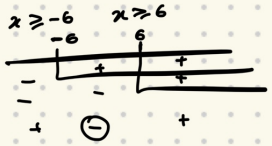
$36-x^2 \geq 0 \rightarrow$

$x^2-36 \geq 0$

$x^2 = 36$

$x_1, x_2 = 6, -6$

$1(x+6)(x-6) \leq 0$



PERCHÈ FACCIAMO L'INTERSEZIONE?

$E = [6, +\infty) \cap [-6, 6] = \{6\}$

$h(x) = \sqrt{\log_6(x-7) - 2}$

D:

$x - 7 > 0 \rightarrow x > 7$

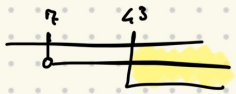
$\log_6(x-7) - 2 \geq 0$

$\log_6(x-7) \geq 2$

$6^{\log_6(x-7)} \geq 6^2$

$x - 7 \geq 36$

$x \geq 43$



$x \geq 43$

$E = [43, +\infty)$

$E = \{x \in \mathbb{R} : x \geq 43\}$

SUCCESSIONI

$a: \mathbb{N} \rightarrow \mathbb{R} \quad a_n$

$\lim_{n\to\infty} a_n = l \in \mathbb{R} \iff \forall n_\varepsilon > 0, \exists n_\varepsilon \in \mathbb{N}:$
$|a_n - l| < \varepsilon \quad \forall n \geq n_\varepsilon$

SE $l = 0$ a_n È DETTA INFINITESIMA

PROPRIETÀ

$\lim_{n\to\infty} a_n = l$

$\lim_{n\to\infty} b_n = m$

$a_n + b_n \rightarrow l + m$

$a_n - b_n \rightarrow l - m$

$a_n b_n \rightarrow lm$

SE $n \neq 0$ $\frac{a_n}{b_n} \rightarrow \frac{l}{m}$

TEOREMA CARABINIERI

$a_n \leq b_n \leq c_n$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = l \Rightarrow \exists \lim_{n\to\infty} b_n = l$

↓ Anche con disuguaglianze strette

$a_n < b_n < c_n$

$a_n = 0, \quad c_n = 1/n, \quad b_n = 1/2n$

MONOTONIA DEL LIMITE

$a_n, b_n \qquad a_n \leq b_n \ \forall n$

$\Downarrow$

$\lim_{n\to\infty} a_n = a \qquad a \leq b$

$\lim_{n\to\infty} b_n = b$

DIM

$\forall \varepsilon > 0 \quad \exists n_\varepsilon, \bar{n}_\varepsilon \in \mathbb{N}: \quad n_\varepsilon = \bar{n}_\varepsilon$

$|a_n - a| < \varepsilon \quad \forall n \geq n_\varepsilon$

$|b_n - b| < \varepsilon \quad \forall n \geq \bar{n}_\varepsilon$

$\forall n \geq n_\varepsilon$

$a - \varepsilon < a_n \leq b_n < b + \varepsilon$

↘ PER $\varepsilon \rightarrow 0$

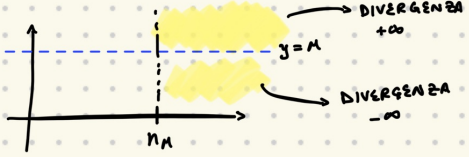
$a \leq b$

SUCCESSIONI DIVERGENTI

DIVERGENZA A $\pm\infty$ $\longrightarrow$ $\{a_n\}_{n\in\mathbb{N}}$

DIVERGE A $+\infty$ SE

$\exists\, n_M \in \mathbb{N}: a_n > M \ \forall n \geq n_M$



ESEMPIO

$$a_n = \frac{n^2}{n+1}$$

$$a_n = \frac{n^2}{n+1} > M \Rightarrow n^2 > M\cdot n + 1$$

$$n^2 > Mn + M$$

$$n^2 - Mn - M > 0$$

$M \rightarrow$ SUPPONIAMO SIA UNA COSTANTE POSITIVA

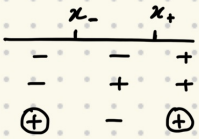
$$M > 0$$

$$x^2 - Mx - M > 0$$

$$x = \frac{M \pm \sqrt{M^2+4M}}{2}$$

$$x_+ = \frac{M + \sqrt{M^2+4M}}{2}$$

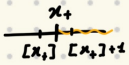
$$x_- = \frac{M - \sqrt{M^2+4M}}{2}$$



CI INTERESSA SOLO QUESTA PERCHÉ n DEVE DIVENTARE GRANDE

$$x > x_+$$

$$n_M = [x_+] + 1$$



↑ PARTE INTERA

$$1 = [1{,}3] \leq 1{,}3 \leq 2$$

SUCCESSIONE REGOLARE

SE $\lim_{n\to\infty} a_n = \ell \in \mathbb{R}$ oppure $\lim_{n\to+\infty} a_n = \pm\infty$

$a_n = (-1)^n \rightarrow$ non regolare

SUCCESSIONI MONOTONE

$a_n \nearrow$ { STRETTAMENTE CRESCENTE $\rightarrow n < m \Rightarrow a_n < a_m$
STRETTAMENTE NON DECRESCENTE $\rightarrow n < m \Rightarrow a_n \leq a_m$

$a_n \searrow$ { STRETTAMENTE DECRESCENTE $\rightarrow n < m \Rightarrow a_n > a_m$
STRETTAMENTE NON CRESCENTE $\rightarrow n < m \Rightarrow a_n \geq a_m$

TEOREMA

LE SUCCESSIONI MONOTONE SONO REGOLARI

$a_n \nearrow \Rightarrow \lim_{n\to\infty} \sup_{n\in\mathbb{N}} \{a_n\}$

$a_n \searrow \Rightarrow \lim_{n\to\infty} \inf_{n\in\mathbb{N}} \{a_n\}$

DIM

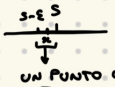
$a_n \nearrow \quad a_1 \leq a_2 \leq a_3 \leq \dots$

$\{a_n\} \subseteq [a_1, +\infty)$

CASO $\sup_n a_n = \ell \in \mathbb{R}$

$\ell = \sup E$: $x \leq \ell \ \forall x \in E$

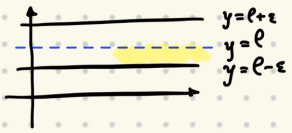
$\forall \varepsilon > 0 \ \exists x \in E: x > \ell - \varepsilon$



↓ UN PUNTO CHE $\in E$

$\forall \varepsilon > 0 \quad \exists n_\varepsilon : \ell - \varepsilon < a_{n_\varepsilon} \leq a_n \leq \ell$

$\forall n \geq n_\varepsilon$ → MONOTONIA



$a_n \downarrow$

$\ell = \inf E \in \mathbb{R}$

$a_1 \geq a_2 \geq a_3 \geq \dots, \{a_n\}$

$\ell = \inf E$

$\ell \leq x \quad \forall x \in E$

$\forall \varepsilon > 0 \quad \exists x \in E : x < \ell + \varepsilon$

$\forall \varepsilon > 0 \ \exists n_\varepsilon : \ell \leq a_n \leq a_{n_\varepsilon} < \ell + \varepsilon$

$\forall n \geq n_\varepsilon$

